

Welcome to NexusMaxEnt



The maximum entropy method is useful for processing DOSY (Diffusion-Ordered Spectroscopy) of samples with diffusion coefficient distributions (e.g., synthetic polymers).

However, there are limited programs capable of processing DOSY data using the maximum entropy method.

Therefore, NexusMaxEnt was developed to enable DOSY processing using the maximum entropy method. Currently, in addition to ADMM KL Mirror Descent, Proximal KL Mirror Descent and ADMM LambertW have been implemented, offering a total of six solvers across CPU and GPU versions.

To promote the widespread use of the maximum entropy method and NexusMaxEnt, I am releasing the CPU version of ADMM KL Mirror Descent for free.

Commercial use and redistribution are permitted.

Other algorithms and GPU versions are paid and not included.

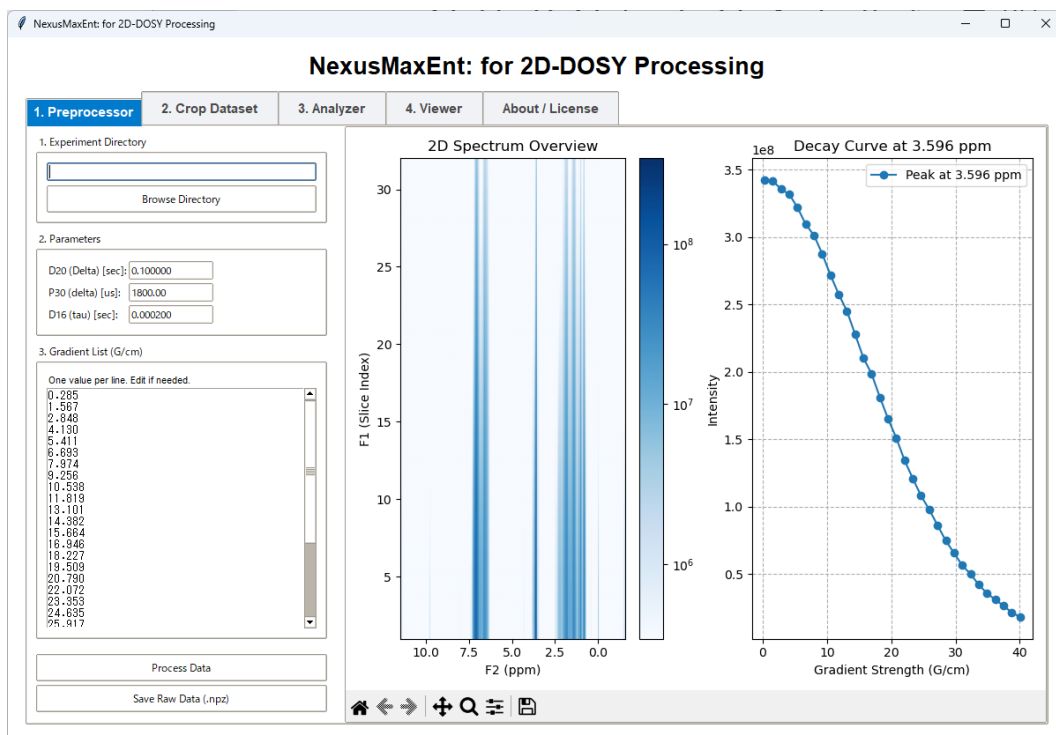
Please check README.md and LICENSE.txt before use.

Please use GitHub Discussions to exchange information and share usage examples with other users.

* This software and guide are provided "as is" for evaluation and research purposes.

Please note that we cannot provide individual technical support or bug fixes.

1. Preprocessor



Specify the directory where the 2rr file is saved from Browse Directory.

Verify that D20, P30, D16, and Gradient List are loaded.

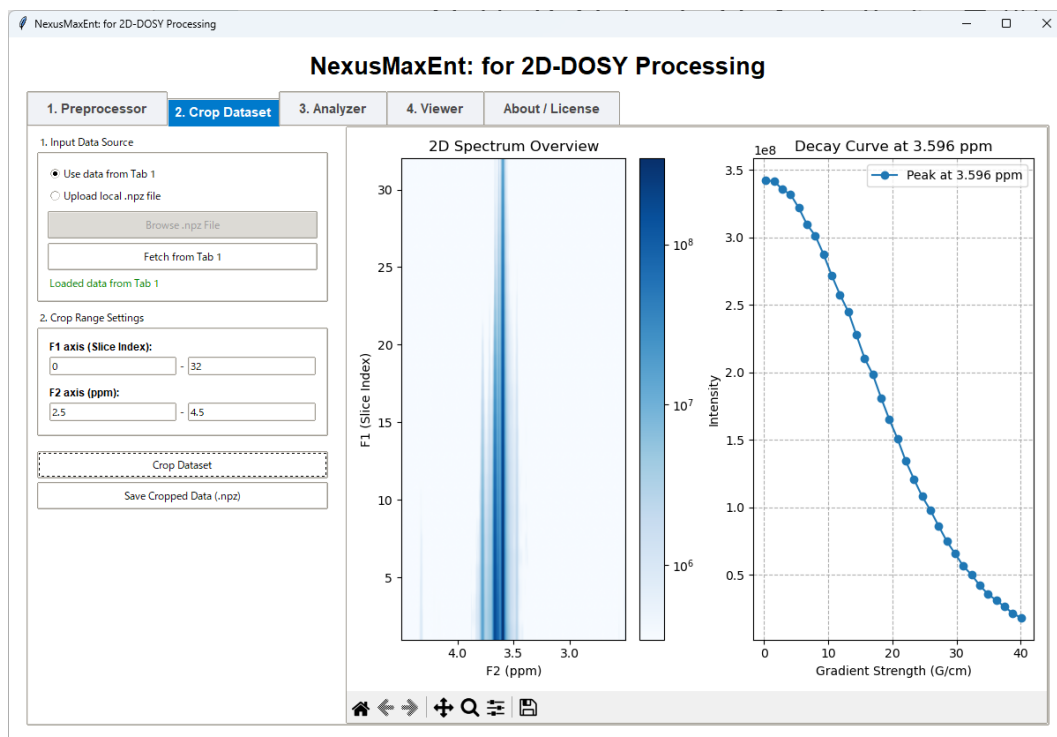
If loading fails, correct it manually.

This is for bpp-type pulse sequences (e.g., ledbpgrp2s). If using a standard STE sequence, halve p30 and set d16 to 0.

See README.md for details.

Pressing Process Data will plot the spectrum and the attenuation of the peak with maximum intensity on the right.

2. Crop Dataset



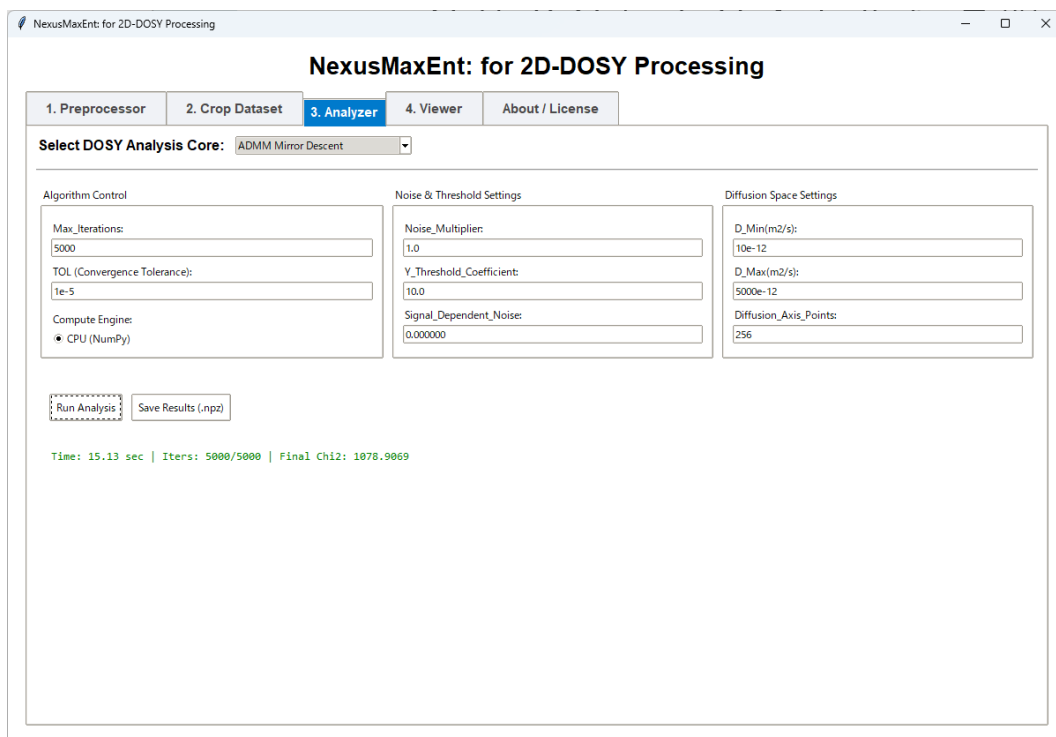
Press "Fetch from Tab1" to transfer data.

To load an npz file saved in Tab1 or a custom-made npz file, select "Browse .npz File" and upload the npz file.

Crop the data as needed.

You can skip this tab if cropping is not necessary.

3. Analyzer



The free version only supports the CPU version of ADMM KL Mirror Descent.

It is basically tuned to work with the default settings.

TOL: This is the threshold at which convergence is considered achieved when the change in the spectrum from the previous iteration is small.

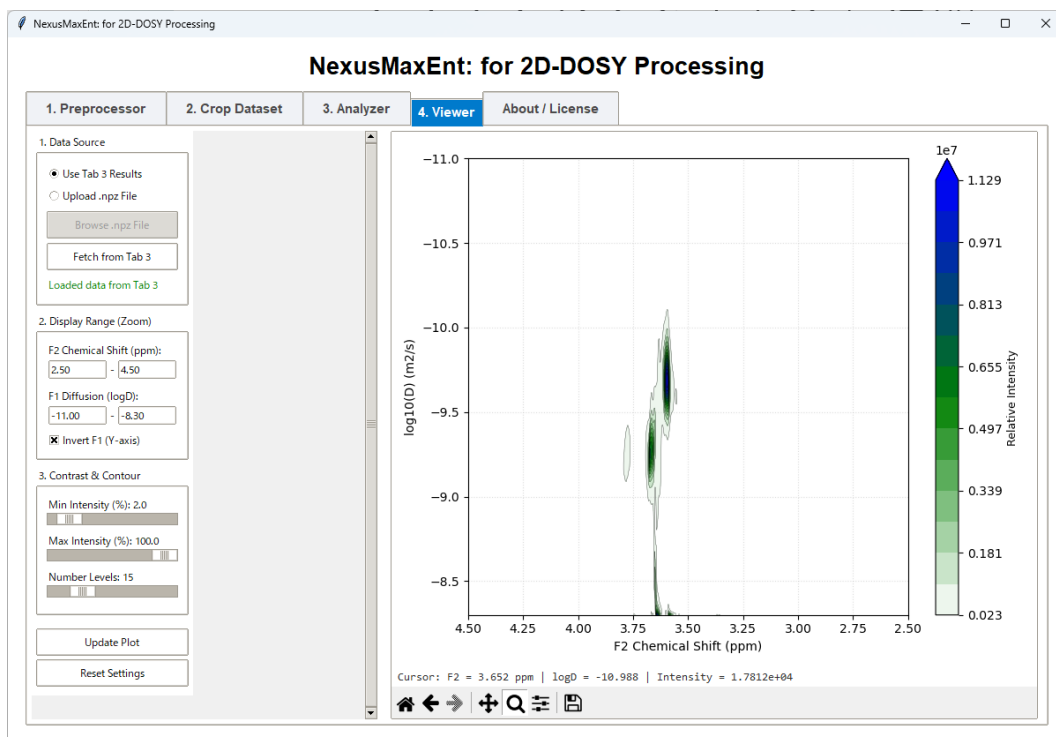
Noise_Multiplier: NexusMaxEnt implements automatic estimation of thermal noise using the MAD method.

This coefficient is a parameter that determines how many times the automatically estimated noise is multiplied.

Y_Threshold_Coefficient: Multiplier applied to the estimated noise floor (via MAD) to define the minimum peak intensity threshold for analysis. (Default: 10.0 or higher is recommended).

Signal_Dependent_Noise: This can reflect noise that depends on peak intensity (such as amplifier stability and phase error). The recommended value is 0, but if overfitting occurs, adjust it to around 0.1% (0.001).

4. Viewer



Fetch from Tab3, or load an analyzed npz file from Browse .npz File.

Because Matplotlib is integrated, you can read chemical shifts and diffusion coefficients using the cursor, and zoom in and out.

The specifications for npz files are published in the README. Please create your own Python code for plotting as needed.